

Aditya Jain

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech. in Computer Science and Engineering (Data Science)

Vellore, Tamil Nadu

2020 – 2025

- **CGPA:** 9.09/10

Delhi Public School

Higher Secondary Education (Class XII)

Meerut, UP

Sep. 2020

- **Result:** 93.4%

Delhi Public School

Secondary Education (Class X)

Meerut, UP

July 2018

- **Result:** 93.6%

EXPERIENCE

AI/ML Engineer

KSMS TECHNOLOGIES SOLUTIONS Pvt. Ltd.

June 2024 – Present

Pune, India

- Developed a **Graph-RAG Agent** integrating **Neo4j** with **SAP ERP** data, enabling **Text-to-SQL/Cypher** capabilities to autonomously execute complex multi-table joins, reducing query retrieval time by **40%**.
- Architected a high-throughput **IDP Pipeline** for engineering RFQs using **PaddleOCR** served via **FastAPI**, implementing geometric alignment and **layout-preserving Markdown** conversion to feed **LLM inference**.
- Engineered a **Hybrid Extraction Logic** combining GenAI reasoning with a **Spatial Template System**, achieving **sub-second processing** for recurring drawing layouts by bypassing redundant LLM calls.
- Built an industrial **Multimodal RAG System** leveraging **LLaMA 3 (8B/70B)**, **MinIO**, and **VectorDB**, deploying **hierarchical retrieval** to achieve machine-level semantic accuracy across maintenance documents.
- Consolidated **Edge AI Perception** on **NVIDIA Jetson (ROS2)**, fusing **GroundingDINO**, **SAM**, and **RGB-D** data for open-vocabulary navigation while managing **YOLOv8** training pipelines.

Junior Data Analyst Intern

Zyvka Global Services

May 2023 – July 2023

Noida, India

- Analyzed market data to identify **growth vectors** and translate complex datasets into **strategic insights**.
- Visualized **financial KPIs** and performance trends to drive data-informed decision-making.

Business Development & Growth Strategy Intern

onTribe

Feb. 2021 – Sep. 2021

Delhi, India

- Executed a **Go-to-Market strategy** that expanded client acquisition and established sustainable **B2B partnerships**.
- Led team **enablement workshops** on operational workflows and lead generation, streamlining the sales funnel.

PROJECTS

Enterprise Graph-RAG & Text-to-SQL for SAP ERP

Neo4j, LangGraph, LLM: Nemotron 253, NVIDIA DGX, FastAPI, Azure SSO

November 2025 – Present

- Modeled complex SAP ERP schemas into Neo4j and built an end-to-end Graph-RAG backend, used daily by **20+ supply chain analysts** to query deeply joined enterprise datasets.
- Engineered a Text-to-SQL/Cypher orchestration layer on on-prem NVIDIA DGX systems, allowing natural-language queries to execute autonomous multi-table joins with sub-second latency, **reducing report generation time from few hours to <10 mins**.
- Integrated Azure SSO (OIDC/SAML) and hardened FastAPI middleware to support **100+ concurrent requests**, eliminating the need for a dedicated **3-person SQL support team** for ad-hoc queries.

Intelligent Document Processing (IDP) for Engineering RFQs

PaddleOCR, FastAPI, OpenCV, LLM, Python, Docker

Jan. 2026 – Present

- Built a high-throughput OCR pipeline for heterogeneous drawings, automating data entry for **1000+ monthly documents** and reducing manual processing workload by **65% (saving ~100 man-hours/month)**.

- Developed computer vision algorithms for deskewing and segmentation, achieving a **98% field-level accuracy rate** and removing the need for manual validation by junior engineers.
- Implemented a spatial template-matching engine to bypass LLM inference for recurring formats, **cutting cloud compute costs by 40%** while maintaining sub-second processing speeds.

Industrial-Grade Maintenance Chatbot (Sovereign AI)

Sept. 2025 – Oct. 2025

MinIO, VectorDB, On-Prem LLM, NVIDIA DGX, MongoDB, Azure AD

- Architected a fully on-prem RAG backend on NVIDIA DGX, ensuring sovereign data control; currently adopted by **50+ field technicians** across 3 manufacturing sites.
- Orchestrated ingestion from MinIO/MongoDB for domain-grounded troubleshooting, directly contributing to a **20% reduction in machinery downtime** by speeding up error diagnosis.
- Delivered a production-ready chatbot that successfully **deflected 40% of Tier-1 support tickets**, freeing up senior engineers to focus on critical infrastructure tasks.

Edge AI & Perceptual Navigation for Quadruped Robotics

June 2025 – Aug. 2025

ROS2, NVIDIA Jetson, GroundingDINO, SAM2, RealSense, Neo4j

- Deployed zero-shot visual grounding on NVIDIA Jetson within a ROS2 stack, enabling autonomous navigation via text prompts with an 87% target reacquisition rate.
- Integrated SAM2 with RealSense RGB-D to construct a semantic object map in Neo4j, enabling navigation to spatial goal poses with 15cm accuracy.
- Optimized the real-time perception-to-control pipeline, reducing detection latency to 120ms for high-speed obstacle avoidance.

TECHNICAL SKILLS

AI/ML: RAG Pipelines, LLMs, LangChain, Cursor, Microsoft Foundry, VectorDB, PyTorch, PaddleOCR, SAM2, YOLOv8, OpenCV

Languages & Data: Python, SQL, Neo4j (Cypher), MongoDB, MinIO, Java

Frameworks & Backend: FastAPI, Docker, REST APIs, ROS2, MQTT, WebSocket, RTSP/RTMP

Tools & Platforms: Git, GitHub, Figma, NVIDIA Jetson, Linux (Shell)

Management & Strategy: Agile/Scrum, Product Lifecycle Management, Growth Strategy, Technical Documentation

LEADERSHIP & ACHIEVEMENTS

Inter IIT Tech Fest

Dec. 2024

Developer (Swarm Robotics)

IIT Mumbai, India

- * Developed a ROS2-based centralized OS for swarm navigation, enabling real-time communication among autonomous agents via **MQTT/WebSockets**.
- * Implemented graph-based path planning and **multi-agent SLAM** for collaborative mapping, integrating LiDAR and vision-based perception for obstacle avoidance.

Director Of Finance

Feb. 2023 – Feb. 2024

VIT Trekking Club

Vellore, India

- * Managed annual **budget allocation** and secured corporate **sponsorships**, optimizing funds to increase event scale and revenue by **20%**.
- * Enforced **financial compliance** and directed internal audits, streamlining resource management and reporting standards.

Operations Manager

Dec. 2020 – June 2022

Entrepreneurship Cell, VIT

Vellore, India

- * Orchestrated logistics and vendor coordination for large-scale **entrepreneurial summits** (including flagship E-Summit 2022), hosting **500+ delegates**.
- * Optimized **operational workflows** and led the **recruitment drive**, structuring training modules for student volunteers.

CERTIFICATIONS

AI & Robotics:: ROS2 for Beginners (Foxy + Humble), Hands-On AI: Build Your Own GPTs, Agentic AI: Planning and Execution

Cloud & Data Science:: Microsoft Security, Compliance, and Identity Fundamentals, Getting Started with Enterprise Data Science, Journey to Cloud

Business & Product:: Product Management: Building a Product Roadmap, Digital Marketing Foundations